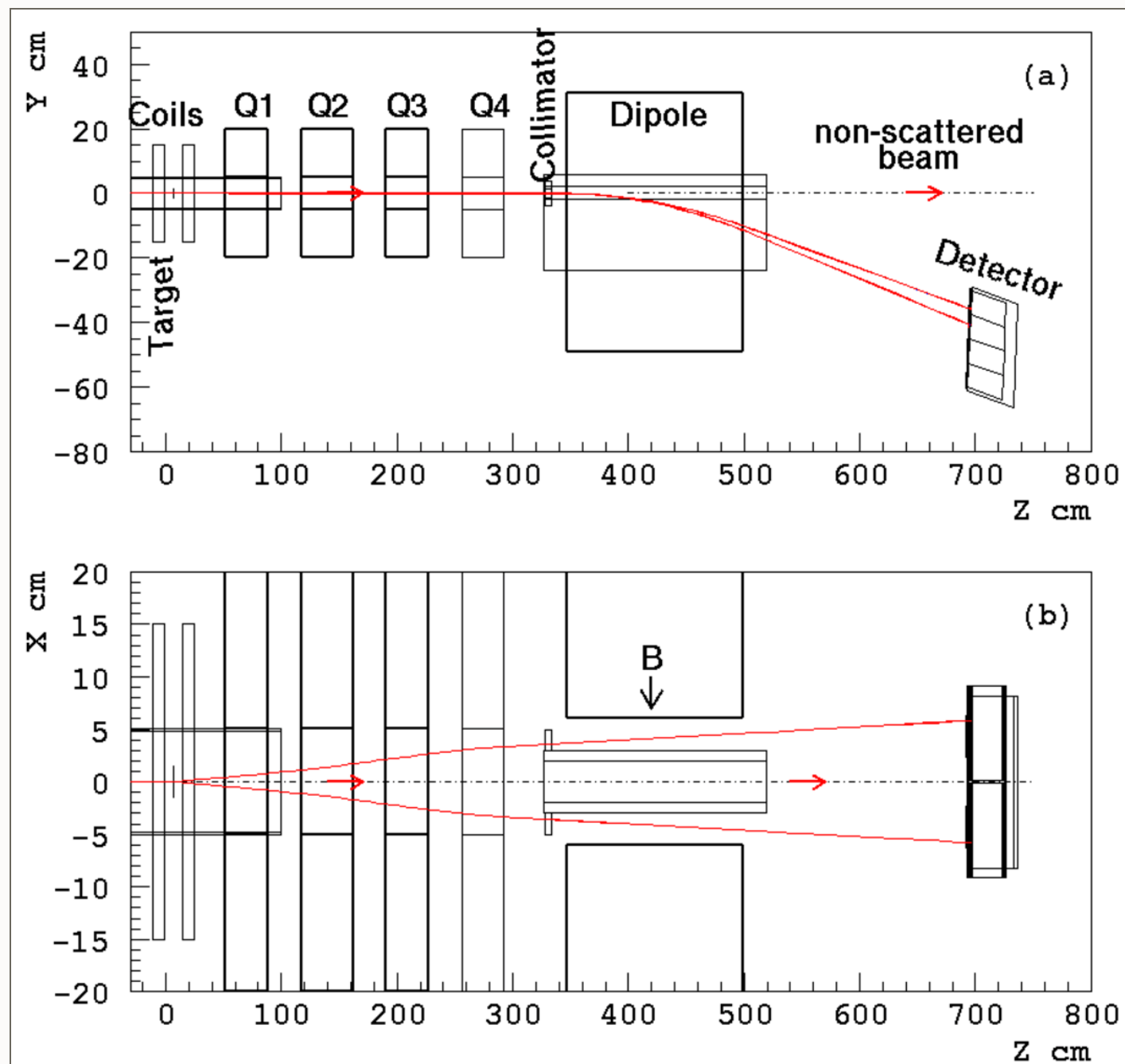


Møller Polarimeter Control Screens

JTabs / MyMenu paths

8 January 2016

The Møller Polarimeter Layout



The Møller Polarimeter Combo Screen

MyMenu → JTabs → Beam Setup → Hall A → Moller → Combo

Hall A Moller Polarimeter Control

BEAMLINE H-COILS TARGET MQO1H02 MQM1H02 MQO1H03 MQO1H03A MMA1H01 DETECTOR

MOELLER ON

MOELLER OFF

Mizar Mode

Beam Sync

Pulse Mode

Normal Mode

Master Mode

Beam Off

Pulse Width Settings

100.00 us

Duty Factor 0.00

Status

Init

Beam structure changes from this screen only take effect if the Master Mode is Expert. If Master Mode is Normal these controls are disabled

Solenoid Status Paused

Solenoid Field -0.000 Tesla

Superconducting Magnet 0.004 kG

PS Comp Gauss Temp

Target Motion Control

LINEAR

Current States

Position 0.000 mm

Position -0.199 deg

Set Linear Position

Right Limit

Park

1 2 3 4

Set Rotary Position

-L

Up Limit

C

Center

+L

Down Limit

0.000 0.050

Jog

Position Input

0.000 -0.200

Motor Status

MAGNET NAME	HYSTERESIS LOOP SWITCH	INTEGRATED FIELD SET POINT (Gauss-Cm)	CURRENT SETPOINT (amps)	CURRENT READBACK (amps)	SET Ramp Rate Amps/sec	On S A T	On S A T	POWER On/Off CONTROLS	PLC RESET
MQO1H02	On Off	0.000	0.000	0.40	6.0	POS	On Off	On Off	RESET
MQM1H02	On Off	246.000	0.000	-0.73	6.0	NEG	On Off	On Off	
MQO1H03	On Off	6763.33000	39.9999	0.000	Positive	Expert	On Off	On Off	
MQO1H03A	On Off	-221.000	0.000	0.00	6.0	POS	On Off	On Off	

MAGNET NAME	HYSTERESIS LOOP SWITCH	INTEGRATED FIELD SET POINT (Gauss-Cm)	INTEGRATED FIELD SET POINT (Gauss-Cm)	CURRENT READBACKS (Amps)	MAIN POWER CONTROLS	
MMA1H01	On Off	1 1964856.00	2 0.000	3 440.000	0.000 0.000	On Off/Reset

MQO1H02

Magnet Current

1.5 0.8 0.6 0.4 0.2 0

0 Minutes

MQM1H02

Magnet Current

0.5 0.4 0.3 0.2 0.1 0

0 Minutes

MQO1H03

Magnet Current

0.8 0.6 0.4 0.2 0

0 Minutes

MQO1H03A

Magnet Current

0.8 0.6 0.4 0.2 0

0 Minutes

MMA1H01

Magnet Current

0.8 0.6 0.4 0.2 0

0 Minutes

Strip Tool

The Møller Target Control Screen

MyMenu → JTabs → Beam Setup → Hall A → Moller → Combo → Target Motion Control

Hall A High-Field Moller Target Motion Control

Linear

Motor Readback (mm) **0.000** Encoder (v) **3.028** Reset

Position (mm)	0	43.444	80.265	117.483	155.298
Encoder (v)	3.025	2.364	1.813	1.251	0.680

Retracted **Park** T1 T2 T3 T4 Extended

P T1 T2 T3 T4

0.000

Position Input (mm)

0.000

< 0.000 >

Jog (mm)

Move

STOP

SM

Rotary

Motor Readback (degree) **-0.199** Encoder (v) **4.510**

Position (degree)	-8.426	-0.200	12.865
Encoder (v)	4.150	4.558	5.206

-Limit **C** **+Limit**

-L C +L

-0.200

Position Input (degree)

0.050

< 0.050 >

Jog (degree)

Move

STOP

SM

The Møller Magnet Temperature Control

MyMenu → JTabs → Beam Setup → Hall A → Moller → Combo → Temp

H
A
C

Moller Polarimeter
Temperature Monitor

Cryocooler (T1)	290.87	K
Magnet (T2)	308.09	K
Magnet Lead #1 (T6)	297.63	K
Magnet Lead #2 (T7)	298.22	K

ID

,MODEL218S,LSB18I,041813

Command

Reset

Reset

IOC Heartbeat

The Møller Magnet Power Supply Screen

MyMenu → JTabs → Beam Setup → Hall A → Moller → Combo → PS

H
A
C

Moller Polarimeter American Magnetics 430

Persistence Heater Control

ON

OFF

Ramp State

Paused

Errors

0,"No

Setpoint

Readback

Current Target

0.000

0.000

A

Current

-0.000

A

Voltage

0.020

V

Field

-0.000

Tesla

Field Units

T

kG

Tesla

Command

Reset Controller

Reset

IOC Heartbeat

The Møller Magnet Compressor Control

MyMenu → JTabs → Beam Setup → Hall A → Moller → Combo → Comp

H
A
C

Moller Polarimeter
Cryomech CP1110

Compressor Mins

144689

mins

Bytes

High Side Pressure

205.2

PSI

Low Side Pressure

204.6

PSI

Input Water Temp

22.0

C

Output Water Temp

22.0

C

Hash Code [Index]

0x0

[1]

Response

0

IOC Heartbeat

The Møller Magnet Gaussmeter Screen

MyMenu → JTabs → Beam Setup → Hall A → Moller → Combo → Gauss

H
A
C

Moller Polarimeter
Gaussmeter

Field

0.004

kG

ID

MODEL450,0,020403

Command

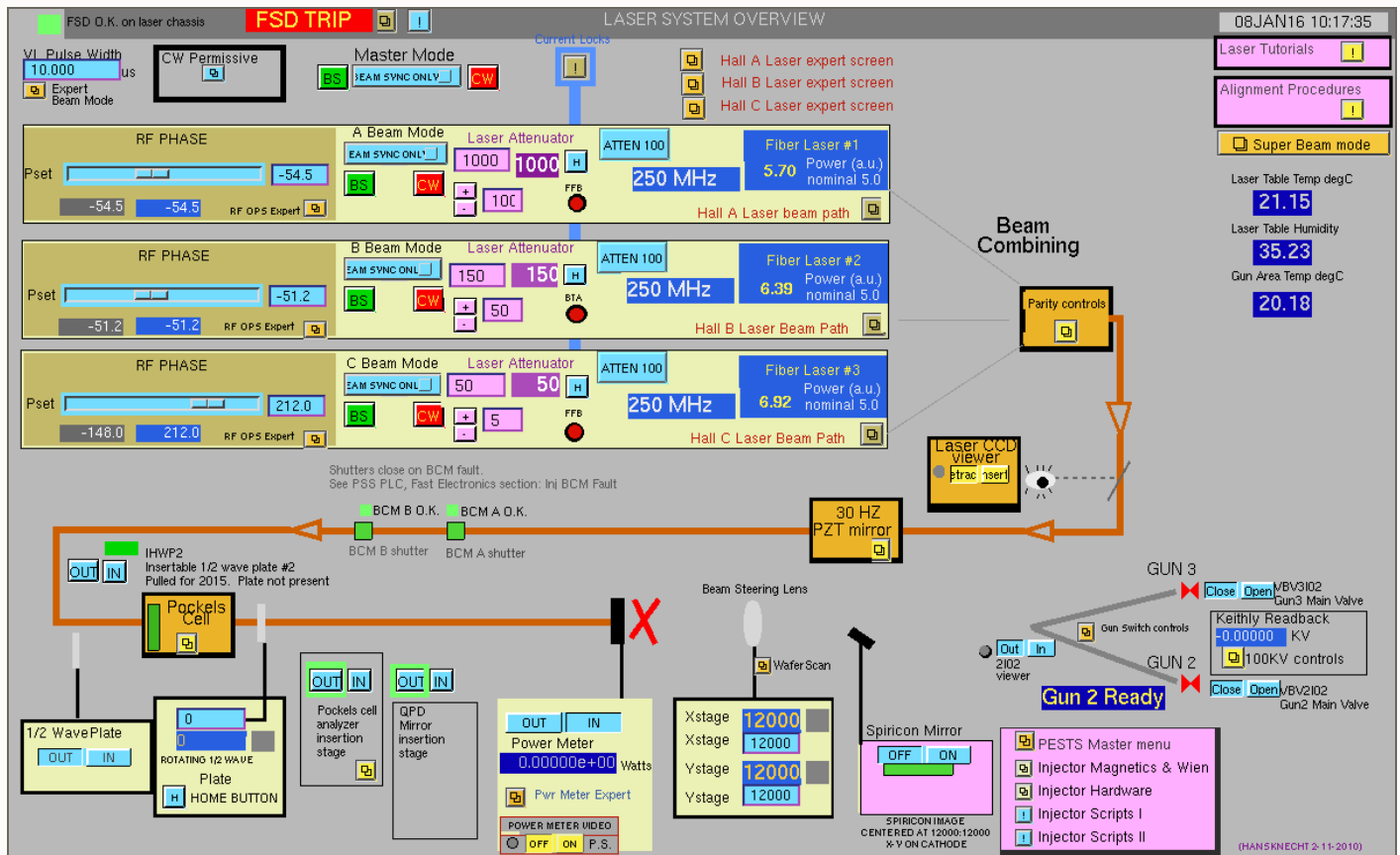
Reset

Reset

IOC Heartbeat

PGun General Tools Screen

MyMenu → JTabs → Injector → Gun/Laser → PGUN Main



Helicity Control Board Screen

MyMenu → JTabs → Injector → Parity → Helicity Control

Helicity Control Board

On configuration change run Configure Injector Alarm Limits script -> !

CONTROL

MONITOR

Mode	Free Clock	Free Clock
1 T-SETTLE Input 1	100 usec	100.00 usec
2 T-SETTLE Input 2		
1 T-STABLE Input 1		4066.65 usec
2 T-STABLE Input 2	4066.65 usec	
Helicity Pattern	Quartet	Quartet
Reporting Delay	16 windows	16 windows
Helicity Board Frequency (Hz)		240.00

Quick Reference

User Guide !

Hardware Rev. MONTH 2 DAY 28 YEAR 14

Spin Control Screen

MyMenu → JTabs → Injector → Spin → Spin Control

CEBAF Expert Spin Controls

Sets Spin Flipper

Longitudinal

Vertical Wien (MWF1104)

Wien Filter Magnet

-0.011

0.989

0.000

Bdl (Gcm)

On

Off

Wien Filter High Voltage

OFF

ON

Power Switch for High Voltage

POSITIVE

Negative

Polarity

*Turn power off before switching polarity

V-WIEN VOLTAGE SET (0 TO 15000V) FOR 0 TO 65535 COUNTS

0

COUNTS SET

0

COUNTS READ

0.00000

VOLTS OUT

Cal Factors & Spin Equation

VWienC1

0.000e+00

VWienC2

1.834e-03

VWienC3

0.000e+00

VWienC4

1.000e+00

VWienC5

-1.020e+01

VWienC6

5.365e-02

VWienAngle = VWienC4 * S

(VWienC1 + VWienC2*V + VWienC3*V^2)

V = C1068_QDAC15

S = IGL100ODAC16_4

VWienAngle = 0.00

Solenoids (MFG1104A/B)

A

0.000

144.958

7.340

Bdl (Gcm)

On

Off

B

0.000

-21.362

9.180

Bdl (Gcm)

On

Off

cal_FGA

4.505e-02

cal_FGB

4.505e-02

Phi_FGA = cal_FGA * MFG1104A.BDL

Phi_FGB = cal_FGB * MFG1104B.BDL

Phi_FG = Phi_FGA + Phi_FGB

Phi_FG = 0.744

Sets Wien Angle

Horizontal Wien (MWF0102)

Wien Filter Magnet

3.965

-0.059

1052.400

Bdl (Gcm)

On

Off

Wien Filter High Voltage

OFF

ON

Power Switch for High Voltage

POSITIVE

Negative

Polarity

*Turn power off before switching polarity

H-WIEN VOLTAGE SET (0 TO 15000V) FOR 0 TO 65535 COUNTS

21354

COUNTS SET

21354

COUNTS READ

3.25841

VOLTS OUT

Cal Factors & Spin Equation

HWienC1

0.000e+00

HWienC2

1.714e-03

HWienC3

0.000e+00

HWienC4

1.000e+00

HWienC5

1.250e+00

HWienC6

4.946e-02

HWienAngle = HWienC4 * S

(HWienC1 + HWienC2*V + HWienC3*V^2)

V = C1068_QDAC16

S = IGL100OD16_9

HWienAngle = 36.60

* To change polarity:

(1) Set the DAC to 0.0.

(2) Turn the power switch off and wait 10 seconds until the HV readbacks below fall to near zero.

0.06

0.00

(3) Select the polarity and turn power on.

Spin setting calculation

Tcl Bpm screens

1.75

1.69

(3) Select the polarity and turn power on.

Spin Calculator Screen

MyMenu → JTabs → Injector → Spin → Spin Calculator

Current Settings		Calculator	
Injector Energy (MeV)	123.00		
North Linac Energy (MeV)	1090.00		
South Linac Energy (MeV)	1090.00		
	Pass	P ² at Hall	
☐ Hall A	5	0.531	
☐ Hall B	1	0.981	
☐ Hall C	3	0.028	
Current Wien Setting		36.601	
Flipper Direction		Longitudinal	
Exit		Help	

Møller Target FSD Screen

MyMenu → JTabs → Operation → FSD → FSD Overview → 1H001

1H001 () (Normal Card)

UM	S	R	G	F	
					Watchdog Timer - 1H001
					BLM ILM1P01 (Compton)
					BLM ILM1P02 (Compton)
					BLM ILM1P03 (Compton)
					BLM ILM1P04 (Compton)
					Vacuum - shield wall to dump
					Vacuum - compton chicane
					Moeller Target
					1H003 Tree

UM = Unmask / Mask

S = Mask Set (grey=unmasked yellow=masked)

R = Mask Readback (grey=unmasked yellow=masked)

G = Gold Mask (grey=unmasked yellow=masked)

F = Latched Fsd Fault (green=good red=fault)

Bit #0 = Top Bit #15 = Bottom

Fault Trip Reset

Mask Set 65344

Mask Readback 65344

Gold Mask 65344

Latched Fault 32636

Mask Enabled 1

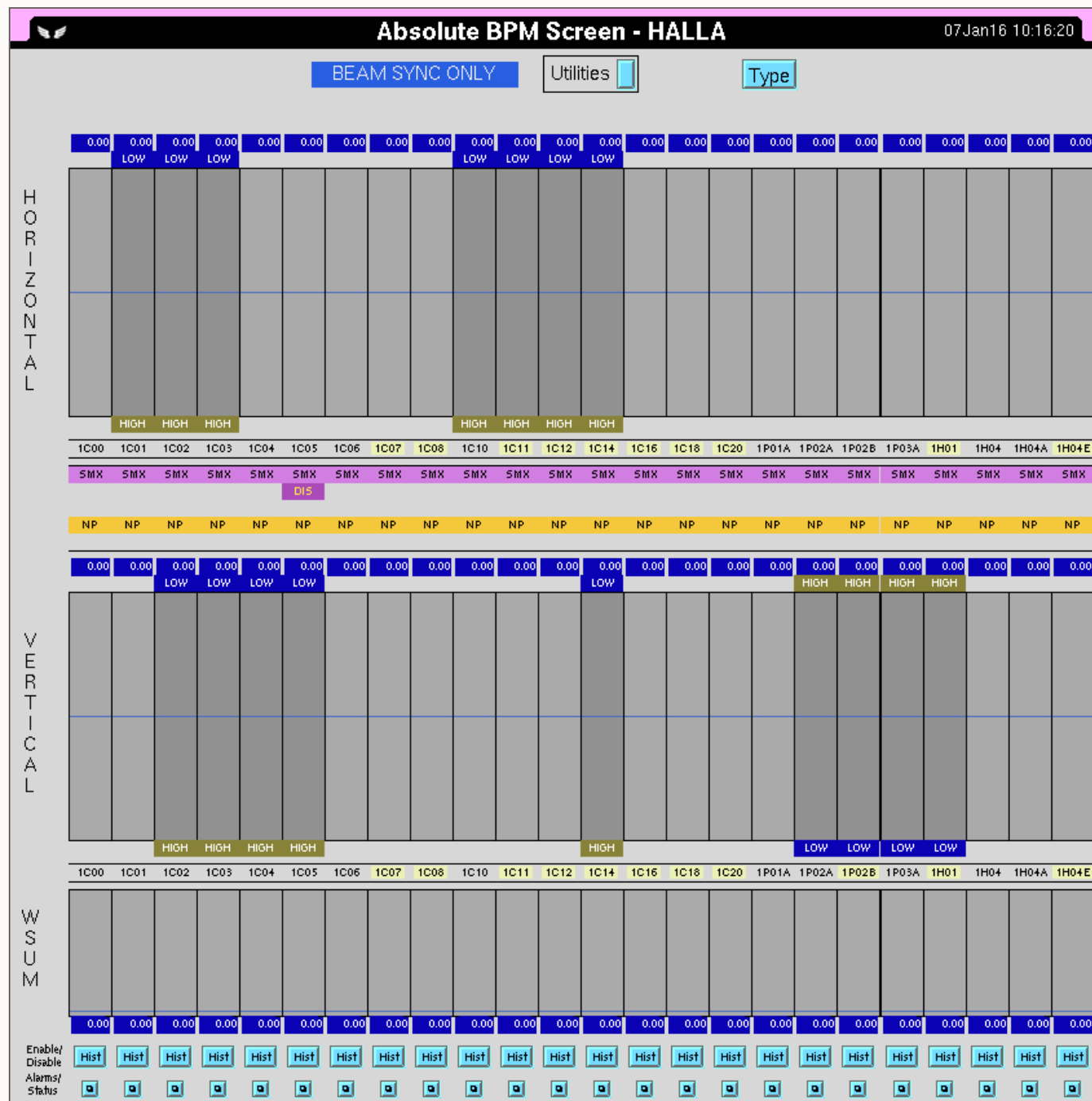
BPM Summary Screen

MyMenu → JTabs → Operations → BPM Spikes → BPM Offset Summary → Halls → Hall A

Offset Summary																07Jan16 10:15:44
Type	Flags	BPM	X Absolute	Y Absolute	X Relative	Y Relative	X Offset	Y Offset	X Quad Cen.	Y Quad Cen.	X Survey	Y Survey	X RMS	Y RMS	ZPos Time	
SEE		NP	IPM1C00	0.000	0.000	-0.844	-4.053	0.000	0.000	0.000	0.000	3.420	0.120	0.000	0.000	21DEC15 05:07:31
SEE		NP	IPM1C01	0.000	0.000	-0.121	-1.785	0.000	0.000	0.000	0.000	0.136	0.071	0.000	0.000	21DEC15 05:07:31
SEE		NP	IPM1C02	0.000	0.000	-0.334	-0.985	0.000	0.000	0.000	0.000	0.280	0.273	0.000	0.000	21DEC15 05:07:31
SEE		NP	IPM1C03	0.000	0.000	-0.700	-1.526	0.000	0.000	0.000	0.000	0.070	0.047	0.000	0.000	21DEC15 05:07:31
SEE		NP	IPM1C04	0.000	0.000	-0.161	-1.191	0.000	0.000	0.000	0.000	0.238	0.165	0.000	0.000	21DEC15 05:07:31
SEE	dlc	NP	IPM1C05	0.000	0.000	-0.156	-1.204	0.000	0.000	0.000	0.000	0.153	0.112	0.000	0.000	20DEC15 20:10:12
SEE		NP	IPM1C06	0.000	0.000	-0.429	0.103	0.000	0.000	0.000	0.000	0.027	0.149	0.000	0.000	21DEC15 05:07:31
SEE		NP	IPM1C07	0.000	0.000	0.085	0.756	0.000	0.000	0.000	0.000	0.098	0.075	0.000	0.000	21DEC15 05:07:30
SEE		NP	IPM1C08	0.000	0.000	0.292	0.748	0.000	0.000	0.000	0.000	-0.001	-0.105	0.000	0.000	21DEC15 05:07:30
SEE		NP	IPM1C10	0.000	0.000	-0.247	-0.278	0.000	0.000	0.000	0.000	0.280	0.079	0.000	0.000	21DEC15 05:07:31
SEE		NP	IPM1C11	0.000	0.000	0.387	-0.467	0.000	0.000	0.000	0.000	0.950	0.450	0.000	0.000	21DEC15 05:07:30
SEE		NP	IPM1C12	0.000	0.000	0.388	-0.568	0.000	0.000	0.000	0.000	0.315	0.000	0.000	0.000	21DEC15 05:07:30
SEE		NP	IPM1C14	0.000	0.000	0.409	-1.111	0.000	0.000	0.000	0.000	0.460	0.053	0.000	0.000	21DEC15 05:07:30
SEE		NP	IPM1C16	0.000	0.000	0.234	0.450	0.000	0.000	0.000	0.000	0.250	-0.100	0.000	0.000	21DEC15 05:07:30
SEE		NP	IPM1C18	0.000	0.000	0.071	0.391	0.000	0.000	0.000	0.000	0.300	0.000	0.000	0.000	21DEC15 05:07:30
SEE		NP	IPM1C20	0.000	0.000	-0.012	0.550	0.000	0.000	0.000	0.000	0.250	-0.200	0.000	0.000	21DEC15 05:07:30
SEE		NP	IPM1P01A	0.000	0.000	No ZPOS	No ZPOS	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	21DEC15 05:07:31
SEE		NP	IPM1P02A	0.000	0.000	No ZPOS	No ZPOS	0.000	0.000	0.000	0.000	0.380	0.100	0.000	0.000	21DEC15 05:07:31
SEE		NP	IPM1P02B	0.000	0.000	No ZPOS	No ZPOS	0.000	0.000	0.000	0.000	0.270	-0.070	0.000	0.000	21DEC15 05:07:31
SEE		NP	IPM1P03A	0.000	0.000	No ZPOS	No ZPOS	0.000	0.000	0.000	0.000	0.580	2.240	0.000	0.000	21DEC15 05:07:31
SEE		NP	IPM1H01	0.000	0.000	-0.531	0.436	0.000	0.000	0.000	0.000	0.310	0.340	0.000	0.000	21DEC15 05:07:31
SEE		NP	IPM1H04	0.000	0.000	0.835	1.716	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	21DEC15 05:07:31
SEE		NP	IPM1H04A	0.000	0.000	0.370	2.805	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	21DEC15 05:07:31
SEE		NP	IPM1H04E	0.000	0.000	0.695	3.316	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	21DEC15 05:07:31

Flag Key

MyMenu → JTabs → Operations → BPM Spikes → Absolute → Halls → Hall A



Beam Energy Monitor

MyMenu → JTabs → Beam Setup → Hall A → Optics Apps → BEM

Beam Energy Monitor

08Jan16 10:46:55

		dp/p bpm	dp/p corr	dp/p total	Momentum	MMS	Valid
INJ	offon	0.00e+00	0.00e+00	0.00e+00	121.35	123.00	
ARC1	offon	0.00e+00	0.00e+00	0.00e+00	0.00	1213.00	
ARC2	offon	0.00e+00	0.00e+00	0.00e+00	0.00	2303.00	
ARC3	offon	0.00e+00	0.00e+00	0.00e+00	0.00	3393.00	
ARC4	offon	0.00e+00	0.00e+00	0.00e+00	0.00	4483.00	
ARC5	offon	0.00e+00	0.00e+00	0.00e+00	0.00	5573.00	
ARC6	offon	0.00e+00	0.00e+00	0.00e+00	0.00	6663.00	
ARC7	offon	0.00e+00	0.00e+00	0.00e+00	0.00	7753.00	
ARC8	offon	0.00e+00	0.00e+00	0.00e+00	0.00	8843.00	
ARC9	offon	0.00e+00	0.00e+00	0.00e+00	0.00	9933.00	
ARCA	offon	0.00e+00	0.00e+00	0.00e+00	0.00	11023.00	
HALLA	offon	0.00e+00	0.00e+00	0.00e+00	0.00	11023.00	
HALLD	offon	0.00e+00	0.00e+00	0.00e+00	12062.11	12113.00	

Master switch

offon

Heart beat

Hall A Combo

MyMenu → JTabs → Operations → Wall Menu → INJ/Halls → Hall A

HALL A
Pass 5

Off
10986.560 MeV

MPS BCM
0.00

1H04 BCM
0 uA

Masks
Compton Valves

Horizontal
Vertical

Low

Low

POS

High

High

1C08

1C00 1C02 1C04 1C06 1C08 1C10 1C12 1C14 1C16 1C18 1P02A 1P03A 1H04 1H04D

Position/Energy MODULATION OFF

IPM1C12
POS X **0.000** POS Y **0.000**

IPM1H01
POS X **0.000** POS Y **0.000**
POF **-0.531** POF **0.436**

IPM1H04A
POS X **0.000** POS Y **0.000**
POF **0.370** POF **2.805**

IPM1H04D
POS X **No ZPOS** POS Y **No ZPOS**

Diffuser Blower

Calorimeter IN

Septum Left Right

Compton Rates
Electron **0**
Photon **0**
Laser Power **0.01**

Striptool

Phone: 5501

Relative Zero Pos Date
21DEC15 05:07:30

FFBA Rel **SEE Norm Ops** **Standby**

OFF **Gain-OK**

Diffuser Setting: **Plate 2 - 30%**

eforman 03/08/2015

Hall A Ion Chambers

MyMenu → JTabs → Operations → RadCon → Ion Chambers → Hall A

Hall A Ion Chambers

Ion Chamber	Rads / Hour	Trip Level
1 (-)	30	2000
2 (-)	0	200
3 (-)	0	300
4 (-)	0	500
5 (-)	0	200
6 (-)	0	210
7 (-)	0	200
8 (-)	0	350

Present Target # **0** **Non**

Beam Current (uA) **0.00**

PLC Health Alarm **ON**

Target Comm Alarm **ON**

Current Comm Alarm **ON**

Software Heartbeat **ON**

Fsd **Auto**

Expert

Faults

Auto Trip Mode **OFF**

Auto Trip Copy Alarm **ON**